Brevan Ellefsen

University of Illinois Urbana-Champaign
bce2@illinois.edu
brevanellefsen.github.io

Research Interests

3-manifolds, essential/measured laminations, taut foliations, pseudo-Anosov flows, hyperbolic geometry, knot theory, Teichmüller theory, essential surfaces, normal and almost-normal surfaces, ergodic theory

Education

University of Illinois Urbana-Champaign — Champaign, Illinois
Ph.D. in Mathematics, expected May 2027; advisor: Nathan Dunfield — 2021–present

Brigham Young University — Provo, Utah
B.S. in Mathematics — 2017-2020

Research Appointment

SLMath Program Associate — Berkeley, California
Topological and Geometric Structures in Low Dimensions — March 16–April 24, 2026

Publications and Preprints

1. **Brevan Ellefsen**. *Growth Rate of Essential Surfaces via Measured Laminations*. 2026. arXiv:2608.08833 [math.GT].

Research in Progress

- **Least-Weight Laminations.** Developing a normal-surface theory of least-weight representatives of essential measured laminations.
- **Volume Bounds on Growth.** Investigating bounds on measured-lamination-space dimension, and hence essential-surface growth, in terms of hyperbolic volume.
- **Minimal Laminations.** Studying when essential laminations admit representatives whose leaves are minimal for a suitable ambient metric.

Invited Research Talks

- **Tsinghua University Geometry, Group Theory, and Topology seminar** — Online, November 10, 2026
  Surface Counts and an Approach to ML vis Measured Laminations
- **Shanghai Institute for Mathematics and Interdisciplinary Sciences (SIMIS)** — Online (two-part minicourse), October 7–8, 2026
  Surface Counts and an Approach to ML via Measured Laminations
- **University of Wisconsin–Madison** — Madison, Wisconsin, September 16, 2026
  Growth Rate of Essential Surfaces via Measured Laminations
- **AMS Fall Central Sectional Meeting – Special Session on Foliations and 3-Manifolds** — St. Louis, Missouri, October 18–19, 2025
  Growth Rate of Essential Surfaces via Measured Laminations

Contributed Research Talks

- **Topological and Geometric Structures in Low Dimensions** — SLMath
  Growth Rate of Essential Surfaces via Measured Laminations — February 5, 2026
- **Topological and Geometric Structures in Low Dimensions (Summer Workshop)** — SLMath
  Growth Rate of Essential Surfaces via Measured Laminations — July 28, 2025
- **42nd Workshop in Geometric Topology** — Oberlin College
  Growth Rate of Essential Surfaces via Measured Laminations — June 12–14, 2025
- **Log Cabin Workshop on Foliations** — St. George, Utah
  Universal Circles, Part 3 (expository talk) — January 13–17, 2025

Selected Conferences

Geometric Topology and Its Friends — CIRM, Marseille
Participant — January 25 – February 19, 2026

Computational Problems about 3-Manifolds, Associated Groups and Varieties — Rutgers University
Participant — October 4–5, 2025

The St. Louis Topology Conference: Flows and Foliations in 3-Manifolds — St. Louis, Missouri
Participant — May 17–19, 2024

Honors and Funding

Departmental Fellowship, University of Illinois Urbana-Champaign
Graduate fellowship support — Spring 2027

Departmental Fellowship, University of Illinois Urbana-Champaign
Graduate fellowship support — Fall 2025–Summer 2026

Teaching Experience

University of Illinois Urbana-Champaign
Teaching Assistant: Calculus I (5 semesters), Calculus II (4 semesters), Linear Algebra (2 semesters) — 2021–present

Brigham Young University
Teaching Assistant: Complex Analysis (1 semester) — 2020

Mentoring and Service

Undergraduate Mentor, UIUC Directed Reading Program — University of Illinois Urbana-Champaign, 3 semesters
Mentored undergraduates in dynamics and algebraic topology

Organizer, UIUC Geometric Topology Seminar — University of Illinois Urbana-Champaign, 2025–present
Created and organize an ongoing seminar for junior researchers in geometric topology and geometric group theory